

TOPIC CODE: RT-SWT-005

Proactive Bug Discovery via Agentic Exploration

Research Proposal | Group 4

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June 20, 2026



| Problem & GAP

Context

- Repository-level testing requires understanding entire codebases and documentation.
- "Proactive discovery" means finding bugs without existing human reports.
- Current LLMs (GPT-5, Claude) show severe limitations in this specific task.

PRIMARY GAP

Shared Limitation (GAP-S)

State-of-the-art LLMs (Claude-Sonnet-4.5) only achieve a **16.06%** Fail-to-Pass rate in proactive scenarios.

Conclusion: Missing an autonomous agent architecture with real-time terminal interaction.

Evidence Summary

Paper / Tool	Dataset (N)	Primary Metric	Best Result
TestExplora (2026)	2,389 tasks	F2P Rate (%)	16.06% (Baseline)
SWT-Bench (2025)	GitHub Issues	Reproduction Rate	27.3% (Reactive)
MASTEST (2025)	REST APIs	Bug Detection	78.0% (Human-in-loop)
TESTWEAVER (2026)	10 Services	Line Coverage	+16.4% improvement

*Evidence selected from 37 analyzed publications confirming the critical performance gap.

Research Questions (RQ)

RQ1: AGENTIC VS. STATIC

Does Agentic Exploration exceed the 16.06% threshold of static context?

H0: Performance $\leq$ 16.06%

H1: Performance $>$ 16.06%

Threshold: 16.06% (TestExplora, 2026)

Test: Wilcoxon ($\alpha=0.05$)

RQ2: SAMPLE BUDGET

Does increasing budget to $k=3$ push F2P towards the 29.7% ceiling?

H0: $k=5$ improvement $\leq$ Static floor

H1: $k=5$ approaches 29.7%

Threshold: 29.7% (SWEAgent + GPT-5-mini)

Test: Wilcoxon ($\alpha=0.05$)

Experiment Pipeline



- **Input:** Repository source code + Documentation (No issue reports).
- **Feedback:** GPT-5-mini interacts with terminal, reads logs, and fixes suites.
- **Constraints:** Maximum 80 turns per exploration loop.

Dataset & Baseline

Dataset: TestExplora-Lite

Source: Microsoft Open Source (GitHub)

- **N:** 517 samples from 482 repositories.
- **Domain:** Web, Data Science, Dev Tools.
- **Anti-Leakage:** Post-2023 stream data.
- **Status:** Accessible (Dockerized).

Baseline: Plain Context

Source: TestExplora (2026, Microsoft)

- Claude-Sonnet-4.5 static generation.
- Single-step prompt (No agent loop).
- No terminal interaction tools.
- Direct comparison on the same dataset.

Statistical Plan

Primary Test	Wilcoxon signed-rank test (One-tailed)
Significance (α)	0.05
Effect Size	Cliff's delta (Non-parametric)
Repetitions (K)	K=3 independent runs per configuration
Power	≥ 0.80 target (G*Power analysis)

*Non-parametric tests chosen due to bounded [0-100%] and skewed distribution of F2P scores.

| Threats to Validity

Internal Validity

- **Model Drift:** Silent updates to cloud LLMs.
Mitigation: Log model-version and system-fingerprints.
- **Stochasticity:** Randomness in high temperature (0.8).
Mitigation: Report Median and SD over K=3 runs.

External & Construct

- **Generalizability:** Mostly Python-based samples.
Mitigation: Explicitly state scope; propose Java extension later.
- **Metric Depth:** F2P doesn't capture assertion depth.
Mitigation: Use HP and EC as cross-validation metrics.

Expected Contribution

"Shifting from Reactive to Proactive Discovery"

Framework

An autonomous agent architecture capable of real-time terminal exploration and execution feedback loops.

Performance

Breaking the 16.06% SOTA ceiling for proactive repository-level testing in black-box/white-box scenarios.

Timeline & Roles

Week	Activity	Responsibility
5	Proposal Drafting + Docker Environment Setup	PL, DG, LR
6	Final Submission + Instructor Approval	PL, RW
7	Pilot Run (20% sample) + Distribution Check	LR, MS
8	Full Experiment Run + Statistical Testing	LR, MS, RW

Roles: PL (Lead), DG (Data), LR (LLM Runner), MS (Stats), RW (Writer)

Q&A

Thank you for your attention

Backup slides available for evidence details and
prompt templates.

